Machine Learning Project Documentation

Assignment 3

Water Access & Mortality Prediction

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1. Overview

The project investigates the relationship between water access quality and mortality rates across countries, using 2019 data from three internationally recognised sources WHO/UNICEF JMP water service level data, World Bank mortality statistics, and World Bank GDP per capita figures. The central research question is whether a country's water service profile can reliably predict whether that country falls into a high mortality or low mortality category.

The project aligns with two of the United Nations Sustainable Development Goals. SDG 6 (Clean Water and Sanitation) and SDG 3 (Good Health and Well Being). By modelling the link between these two goals using machine learning, the project provides a data driven approach for assessing where improvements in water infrastructure could have the greatest impact on survival rates.

The code covers data loading and merging, exploratory data analysis, feature engineering, five baseline classification models, feature selection via Recursive Feature Elimination, hyperparameter tuning via GridSearchCV, model evaluation including confusion matrices and ROC curves, and a linear regression extension to predict mortality. The best performing model is serialised using joblib for potential deployment.

Dataset	Source	Key Variable	Year
Water Service Levels	WHO/UNICEF JMP	Coverage by service tier (%)	2019
Child Mortality Rate	World Bank	Deaths per 1,000 live births	2019
GDP per Capita	World Bank	USD current prices	2019

Goal of the Refinement Phase & Its Importance in the ML Pipeline

It aims to produce a classifier that is more accurate. In any machine learning pipeline, a first model reveals what the data can offer, but it rarely represents the best achievable performance. Refinement addresses this gap through feature selection, hyperparameter tuning and etc. Without this phase, a model may perform well on training data but fail in deployment due to overfitting, irrelevant features, or poorly chosen parameters. For a project with public health implications reliability is not optional. A model that misclassifies high mortality countries as low-risk has consequences.

Changes Made Compared to the Initial Model

The initial submission relied on a basic dataset pairing water potability indicators with a binary drinking water safety label with limited geographic grounding. Following feedback, the project was substantially redesigned. The new version uses WHO/UNICEF JMP water service tier data, World Bank child mortality statistics, and World Bank GDP per capita figures to provide a meaningful geographic and socioeconomic foundation. Feature engineering was introduced through income group categorisation and a binary high-mortality target variable. Recursive Feature Elimination was applied to reduce the feature set from seven to five variables, removing redundant intermediate water access tiers. GridSearchCV hyperparameter tuning was performed on the three strongest classifiers, and a regression extension was added to predict continuous mortality rates alongside the classification task

2. Data Loading

Three CSV files are loaded into separate pandas DataFrames. The water dataset is reshaped from long to wide format using pivot_table(), with ISO3 and Country as the index and 'Service level' as the column axis, producing one column each: Safely managed service, At least basic, basic service, limited service, unimproved, and surface water.

Dataset Merging

All datasets share a country code as a common. The water wide frame is first merged with the mortality frame on ISO3 using an inner join, then the result is merged with the GDP frame using a second inner join. Rows with a missing target variable (Mortality_Rate_2019) are dropped. The final merged dataset contains 177 country records

Feature Engineering

Two engineered features are added to extend the analytical scope

- Income Group: GDP per capita is seperated into four income tiers: Low Income, Lower-Middle Income, Upper-Middle Income, and High Income.
- High Mortality (binary target): Countries above the median(6.10) are labelled 1 (high mortality); those at or below are labelled 0. The resulting classes are nearly balanced: 90 low mortality and 87 high mortality countries

2.4 Missing Value Strategy

Missing water coverage values which arise because not all service tiers are reported for every country. tHEY are imputed with zero before model training, treating absence of data as absence of measured coverage. GDP per capita missing values are imputed with the column median. Rows are only dropped if the binary mortality target itself is missing to preserve the maximum number of samples for model training.

2.5 Dataset Summary Statistics

Feature	Count	Mean	Std	Min	Median	Max
Safely Managed Water (%)	129	67.26	29.90	6.11	72.22	100.00
At Least Basic (%)	48	81.38	16.90	39.91	87.64	99.97
Unimproved Water (%)	177	5.38	7.54	0.00	1.77	34.48
Surface Water (%)	169	2.42	4.04	0.00	0.27	21.22
Mortality Rate	177	16.84	21.61	0.40	6.10	108.10
GDP per Capita USD	174	14,506	19,906	234	5,920	112,697

The wide standard deviations in mortality rate (21.61) and GDP per capita (19,906) reflect the extreme difference between high income and low income countries, underscoring the importance of including GDP as a feature and using income group as a stratification variable in exploratory analysis.

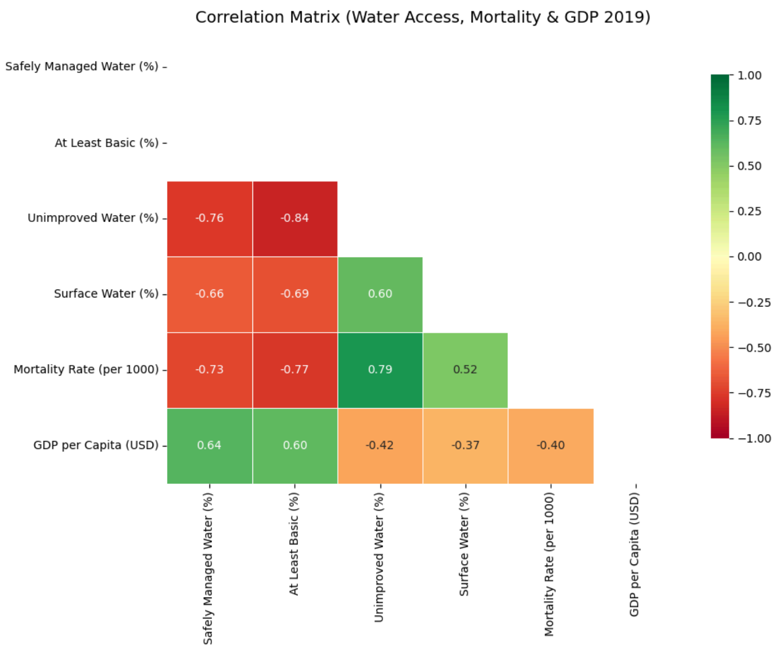
3. Exploratory Data Analysis

3.1 Correlation Analysis

- Safely managed water coverage is strongly negatively correlated with mortality rate (r = −0.73) and moderately positively correlated with GDP per capita (r = +0.64)

- At least basic coverage shows a similarly strong negative correlation with mortality (r = −0.77), confirming that even intermediate access levels matter.
- Unimproved water and surface water access are both positively correlated with mortality (r = +0.79 and r = +0.52 respectively), acting as direct risk indicators
- GDP per capita is negatively correlated with mortality (r = −0.40)

These correlations confirm that water service quality is a stronger predictor of mortality than income level alone.

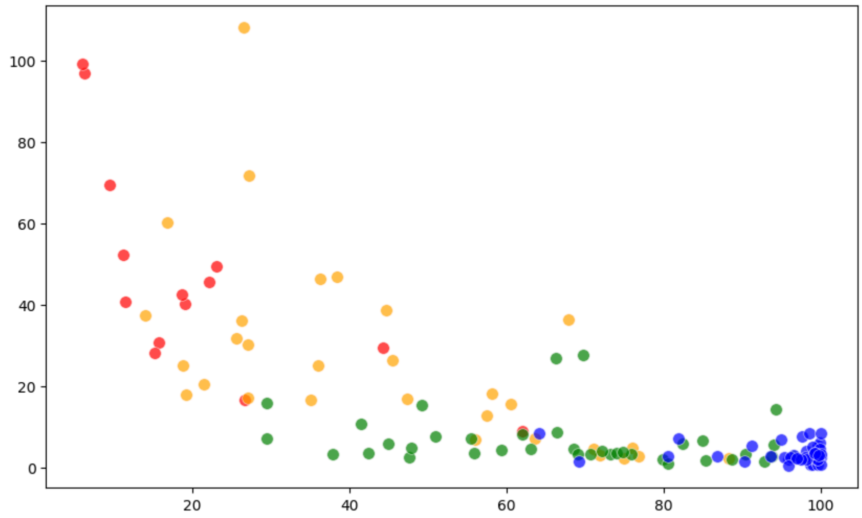


3.2 Scatter Plot: Safe Water Access vs. Mortality by Income Group

A scatter plot with water coverage on the x-axis against mortality on the y-axis, with each data point coloured by income group.

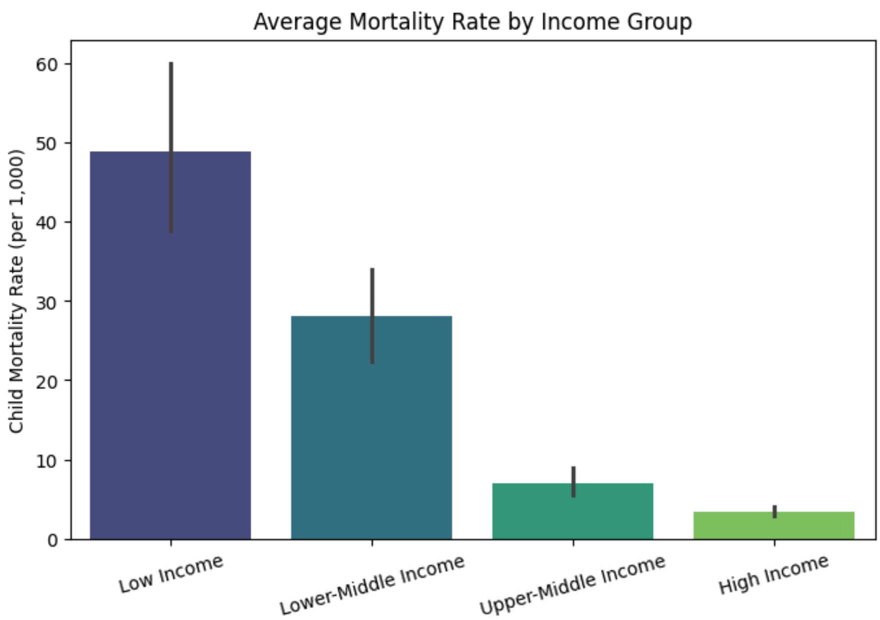
A linear regression trend line (r = −0.725, p < 0.0001)

The visualisation reveals a clear downward trend: countries with higher safely managed water coverage consistently show lower mortality rates. Low income countries (red) cluster in the upper left. Low coverage result in high mortality, high coverage results in low mortality.



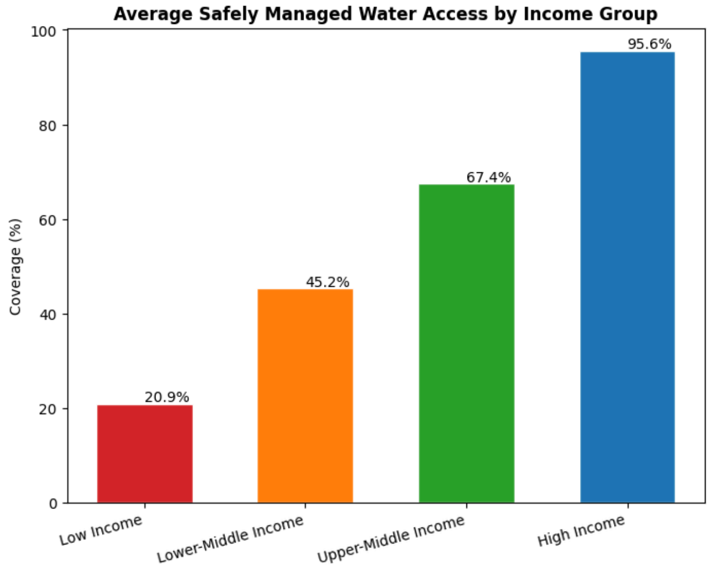
3.3 Average Mortality Rate by Income Group

The bar chart shows the mean mortality rate per income group. Low income countries average approximately 50 deaths per 1,000 births (more than ten times the high-income country average of around 4) Income income group is a useful point for development context but not sufficient on its own to explain within group variance



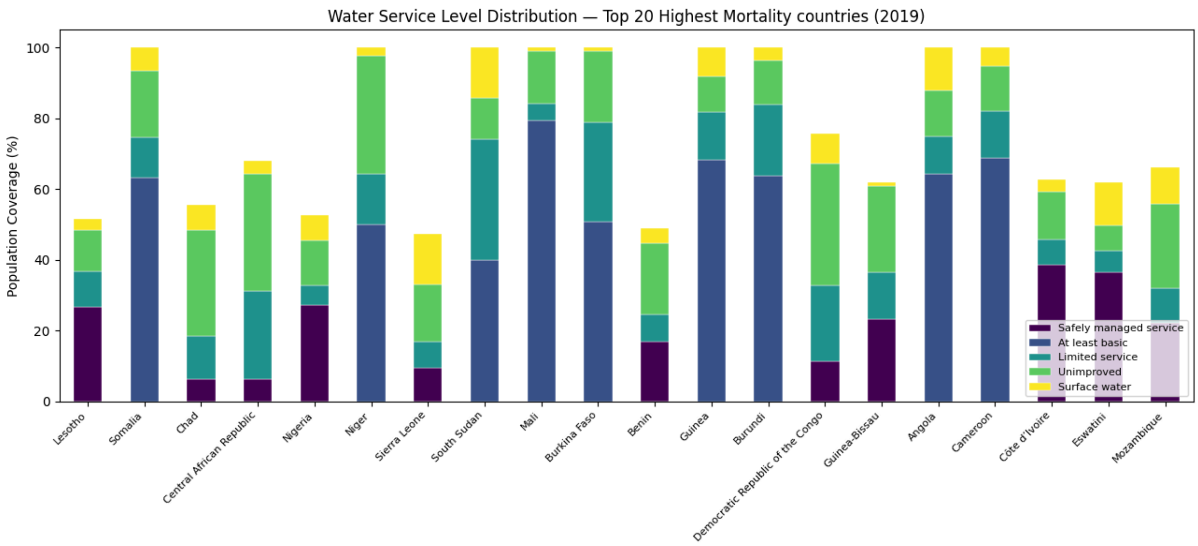
3.4 Average Safely Managed Water Coverage by Income Group

The following shows mean safely managed water coverage by income tier. The high coverage in high-income countries and the 20% value for low-income countries illustrate the structural gap that SDG's want to close.



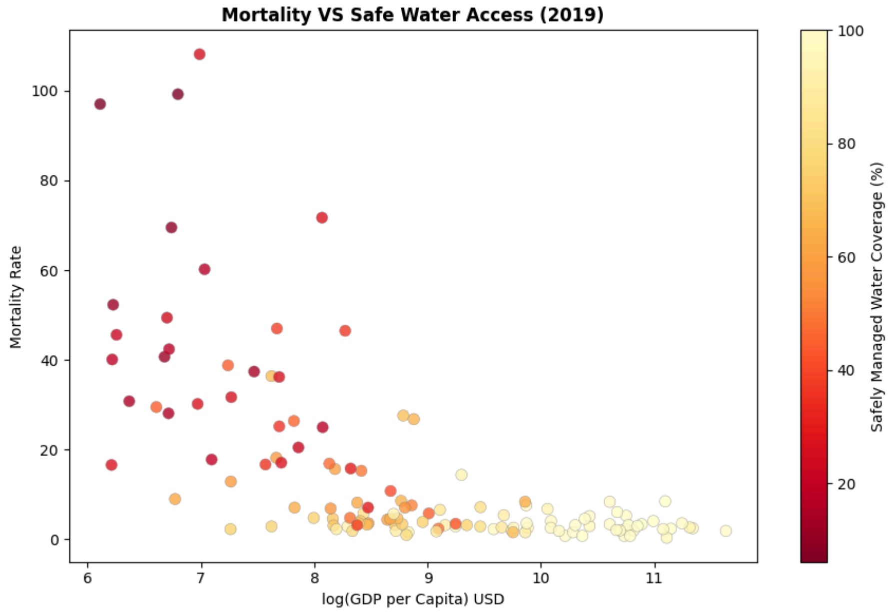
3.5 Water Service Level Distribution in the Highest Mortality Countries

The chart shows the water service tiers for the 20 countries with the highest mortality rates in 2019. Countries such as South Sudan, Angola and Guinea show large proportions of surface water and unimproved access alongside minimal safely managed service



3.6 GDP vs. Mortality with Water Coverage Overlay

The gradient reveals that even at similar income levels, countries with higher water coverage (lightyellow) tend to cluster at lower mortality values, suggesting that water infrastructure improvements have mortality benefits that are partially independent of income.



4. Feature Preparation & Selection

4.1 Initial Feature Set

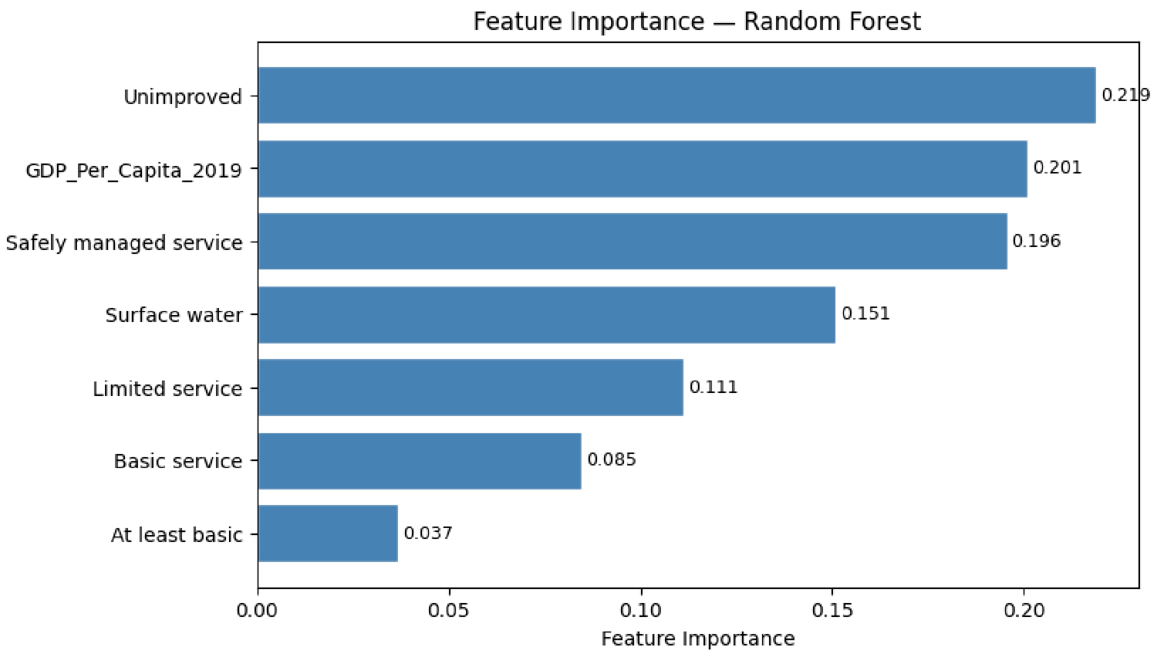
Seven features are passed to the machine learning models: the six water service coverage columns (Safely managed service, At least basic, Basic service, Limited service, Unimproved, Surface water) and GDP per capita. The ISO3 country code and Country name are excluded as non predictive identifiers.

4.2 Recursive Feature Elimination (RFE)

RFE is applied with a Random Forest estimator set to retain five features. The subset is:

- Safely managed service
- Limited service
- Unimproved
- Surface water
- GDP_Per_Capita_2019

The two excluded features (at least basic and basic service) had the lowest Random Forest importance scores (0.037 and 0.085), meaning their predictive signal is captured by the safer and riskier tiers. The top three features by importance are Unimproved, GDP_Per_Capita_2019, and Safely managed service.



4.3 Scaling

All features are standardised using StandardScaler (zero mean, unit variance) before training. The scaler is fitted exclusively on the training set and applied to the test set to prevent data leakage. A separate scaler is fitted for the RFE reduced feature set used in hyperparameter tuning and final evaluation.

4.4 Train Test Split

The 177 ML ready rows are split 80/20 (141train, 36 test) using stratified sampling on the binary mortality target. Stratification ensures both splits maintain the approximately 50/50 class balance present in the full dataset.

5. Model Training & Evaluation

5.1 Baseline Models

Five classifiers are trained on the feature scaled training set as baselines, without hyperparameter tuning. Results on the test set are summarised below. Untuned

Model	Test Accuracy	Weighted F1
Random Forest	0.8333	0.8333
Logistic Regression	0.7500	0.7483

Decision Tree	0.7500	0.7498
K-Nearest Neighbours	0.7222	0.7214
Gradient Boosting	0.7222	0.7222

Random Forest leads with 83.3% accuracy and a weighted F1 of 0.833.(resistance to overfitting on the small dataset) Logistic Regression and decision tree tie at 75.0%, while KNN and Gradient Boosting trail at 72.2%.

5.2 Cross Validation

Stratified cross validation is performed on the training set for all models to obtain more reliable performance estimates than a single train-test split allows.

Model	CV Mean Accuracy	CV Std
Random Forest	0.8512	0.0409
Logistic Regression	0.8443	0.0468
KNN	0.8441	0.0161
Gradient Boosting	0.8089	0.0465
Decision Tree	0.8015	0.0360

Cross validation confirms Random Forest as the most reliable model .KNN produces the lowest standard deviation (0.016), indicating consistent but slightly lower performance.

5.3 Hyperparameter Tuning

GridSearchCV with 5 fold stratified cross validation is applied to three models using the RFE feature set.

Random Forest

- n_estimators: [50, 100, 200]
- max_depth: [None, 5, 10, 20]
- min_samples_split: [2, 5]
- max_features: ['sqrt', 'log2']

CV accuracy: 0.8372

Gradient Boosting

- n_estimators: [50, 100, 200]
- learning_rate: [0.05, 0.1, 0.2]
- max_depth: [3, 5, 7]

CV accuracy: 0.8300

Logistic Regression

- C (regularisation strength): [0.01, 0.1, 1, 10, 100]
- solver: ['lbfgs', 'liblinear']

CV accuracy: 0.8308. The small C value (strong regularisation) reflects the limited sample size, preventing the linear boundary from overfitting.

5.4 Tuned Model Results & Overfitting Analysis

Model	Baseline Acc	Tuned Acc	Baseline F1	Tuned F1	AUC	CV–Test Gap	Status
Random Forest	0.8333	0.8333	0.8333	0.8333	0.875	0.021	Stable
Logistic Regression	0.7500	0.7778	0.7483	0.7800	0.861	0.064	Mild Overfit
Gradient Boosting	0.7222	0.7500	0.7222	0.7500	0.829	0.080	Overfit

Random Forest remains the best option after tuning and is the only model that generalises without overfitting (CV-to-test gap of just 0.021)

6. Detiled Classification Results

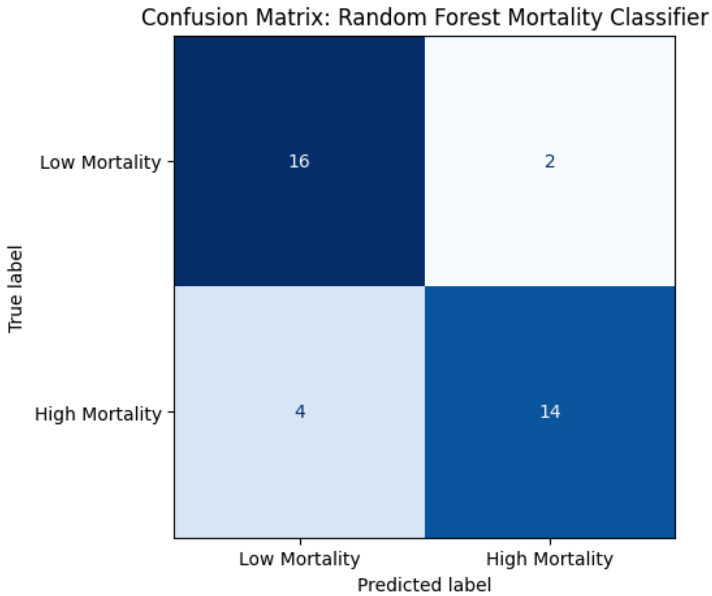
6.1 Random Forest Classification Report

The tuned Random Forest model achieves balanced performance across both classes, with slightly higher precision for detecting high-mortality countries (0.88) than low-mortality countries (0.80).

Class	Precision	Recall	F1-Score	Support
Low Mortality	0.80	0.89	0.84	18
High Mortality	0.88	0.78	0.82	18

Macro Average	0.84	0.83	0.83	36
Weighted Avg	0.84	0.83	0.83	36

The confusion matrix shows 16 true negatives (correctly identified low-mortality countries), 14 true positives (correctly identified high-mortality countries), 2 false positives (low-mortality countries misclassified as high-risk), and 4 false negatives (high-mortality countries missed). The four false negatives represent the more problematic error type for public health policy: countries with genuinely high mortality rates that the model fails to flag.

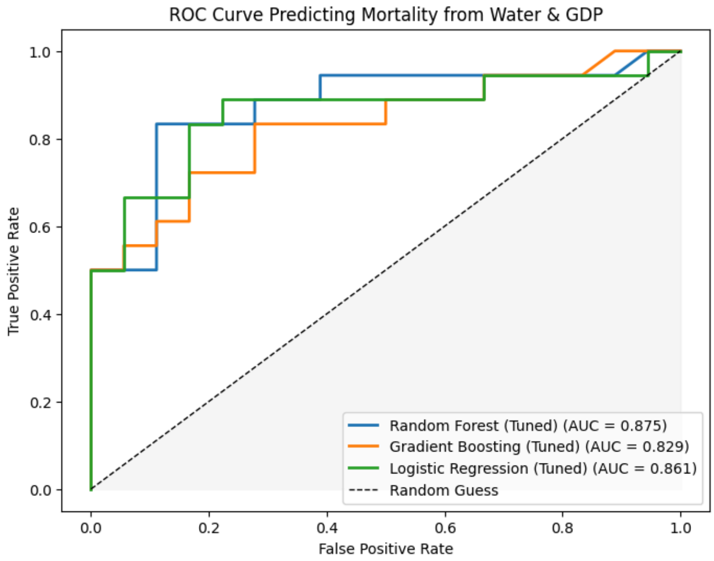


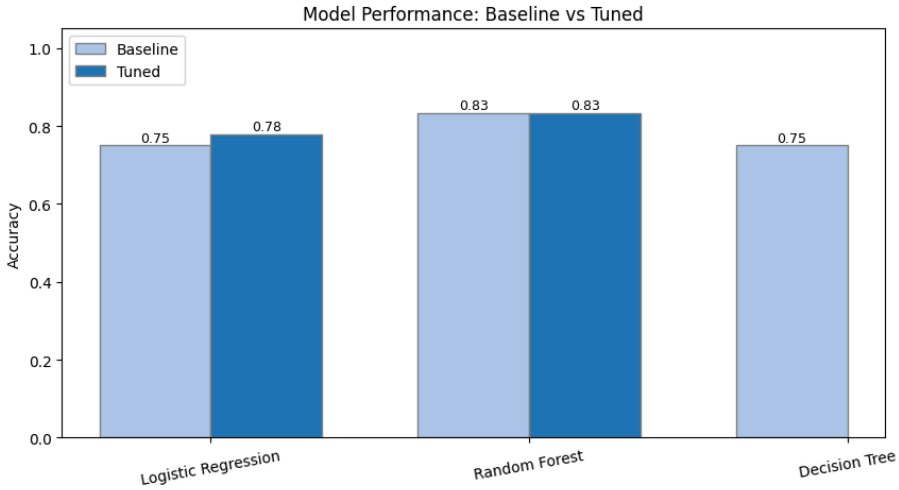
6.2 ROC Curve Analysis

ROC curves are plotted for all three tuned models. The area under the curve (AUC) values are:

- Random Forest: AUC = 0.875   curve that rises steeply before any significant false positive rate
- Logistic Regression: AUC = 0.861 strong linear relationship between water access and mortality.
- Gradient Boosting: AUC = 0.829 consistent with its overfitting pattern on this small dataset.

All three models substantially outperform the random guess baseline (AUC = 0.5), confirming that water service coverage and GDP per capita carry genuine predictive signal for mortality risk classification.





7. Regression Analysis: Predicting Continuous Mortality

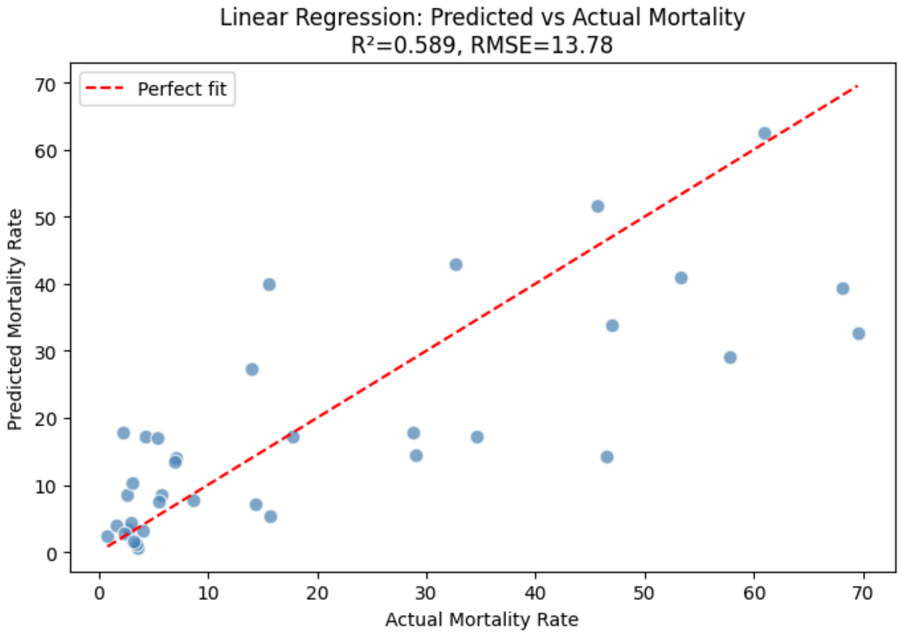
Beyond binary classification, a regression extension is implemented to predict the actual mortality rate as a continuous value which is a more demanding task that tests whether the model can capture the full range of mortality variation

7.1 Linear Regression

A standard Ordinary Least Squares linear regression is trained on the same RFE-selected five-feature set with the same 80/20 split. The model achieves:

- RMSE: 13.78 deaths per 1,000 live births
- R^2: 0.5891 the model explains approximately 59% of the variance in mortality rates

The predicted vs. actual scatter plot shows strong agreement at lower mortality values but increasing scatter at higher values.



7.2 Random Forest Regressor

A Random Forest Regressor is trained on the same setup as a non linear baseline comparison.

- RMSE: 12.84 deaths per 1,000 live births
- R^2: 0.6431

Model	RMSE	R^2	R^2 Improvement
Linear Regression	13.78	0.5891	—
RF Regressor	12.84	0.6431	+0.054

The Random Forest regressor's non-linear boundaries allow it to better model the extreme high-mortality cases that linear regression underestimates. However, even the Random Forest regressor leaves 36% of mortality variance unexplained, reflecting that factors beyond water access and GDP (including healthcare infrastructure, conflict,etc) play important roles in determining mortality rates.

8. Model Serialisation & Deployment Readiness

The best-performing model (tuned Random Forest classifier) and its associated StandardScaler are serialised to disk using joblib, producing two artefacts:

- best_rf_mortality_classifier.joblib — the trained RandomForestClassifier object, containing 100 decision trees fitted on the five RFE-selected scaled features.
- scaler_water_sdg.joblib — the fitted StandardScaler, essential for transforming new input data to the same distribution as the training set before inference.

Correct serialisation is verified by reloading both artefacts and running inference on a synthetic example country profile: safely managed service 45%, at least basic 60%, limited service 10%, unimproved 20%, surface water 5%, GDP per capita $1,500 (Lower-Middle Income). The loaded model returns the predicted mortality class, demonstrating a reproducible prediction pipeline suitable for integration into a API for SDG monitoring.

9. API Integration

The trained Random Forest mortality classification model was deployed using the FastAPI framework.The (best_rf_mortality_classifier.joblib) and scaler (scaler_water_sdg.joblib) are loaded Users can send water access and economic indicators to the API and receive a mortality risk prediction.

The API scales the input features using the saved scaler and then generates a mortality risk prediction using the trained Random Forest classifier.

Security Considerations

Several security mechanisms were incorporated into the deployment design to protect the prediction service and ensure only authorized access.

Authentication and Authorization

The API uses API-key based authentication. Clients must provide a valid API key in the x-api-key request header before accessing the prediction endpoint.

If an invalid key is supplied, the API returns an HTTP 401 Unauthorized response

Data Encryption

In a production environment, the API would be deployed behind HTTPS using TLS encryption. This ensures that all data transmitted between the client and server remains encrypted and protected from interception.

Threat Mitigation

The following measures help reduce security risks:

- Automatic type checking and range validation for all input variables.
- API key authentication.
- Error handling using HTTP exceptions.

For example, percentage based water indicators are restricted to values between 0 and 100, preventing invalid inputs from reaching the model.

Monitoring and Logging

Monitoring and logging are essential for ensuring reliable model operation after deployment.

The following operational metrics can be tracked:

- Prediction request count
- Successful prediction rate
- Failed request rate
- Authentication failures
- API response time

Logging

The API logs:

- Authentication failures
- Request timestamp
- Prediction result
- Prediction confidence
- Runtime errors

10. Conclusions & Key Findings

10.1 Summary of Results

Metric	Value
Best classifier	Random Forest
Test accuracy	83.3%
AUC	0.88
CV–Test gap	0.0039 (stable)
Regression R^2 (Random Forest)	0.64
Regression RMSE (Random Forest)	12.84

10.2 Key Insights

- Water quality predicts mortality better than income alone: the correlation between unimproved water access and mortality (r = +0.79) exceeds that between GDP and mortality (r = −0.40). This suggests that targeted water infrastructure investment may deliver mortality reductions even in countries with limited GDP growth.
- Random Forest consistently outperforms all other tested models on this dataset, with 83.3% accuracy and an AUC of 0.875. Its robustness to overfitting (low 0.021 CV-to-test gap) makes it the recommended model)

- Logistic Regression's competitive AUC (0.861) reflects the essentially linear relationship in the data and indicates that interpretable linear classifiers are viable when transparency is prioritised over marginal accuracy gains.
- The regression extension shows that water access and GDP explain approximately 64% of mortality variance when modelled non-linearly. The remaining 36% is attributable to factors outside the dataset

10.3 Limitations

- Small sample size: 177 countries is a small dataset for machine learning. Cross validation mitigates this risk but cannot fully substitute for larger samples.
- Synthetic zero imputation: Missing water coverage values are imputed as zero, which overstates the proportion of the population without any measured access and may introduce a consistent bias in countries where data collection isnt very good.
- Single year data: All data is from 2019. Time series modelling of how improvements in water access translate into mortality reductions over years would provide stronger causal inference..

10.4 Future Work

- Integrating more wide JMP data (2000–2022) to model lagged effects of water improvements on mortality keeping in mind that correlating mortality and gdp data can be found
- Add conflict and healthcare expenditure features from the World Bank databases.
- Deploy the serialised model as a FastAPI endpoint with country-code lookup, providing a real time monitoring tool.